

WEEK 2 – POWER BI DATA ANALYSIS

Superstore Sales & Profit Analysis Dashboard

1. Introduction

Microsoft Power BI is used to analyze and visualize business data. This project uses the Superstore Sales Dataset to study sales, profit, orders, products, customers, and regions.

2. Objectives

Analyze Superstore sales data.

Clean and organize the dataset.

Create a data model and Date Table.

Calculate important metrics using DAX.

Analyze sales and profit by category and region.

Create an interactive dashboard.

3. Dataset Description

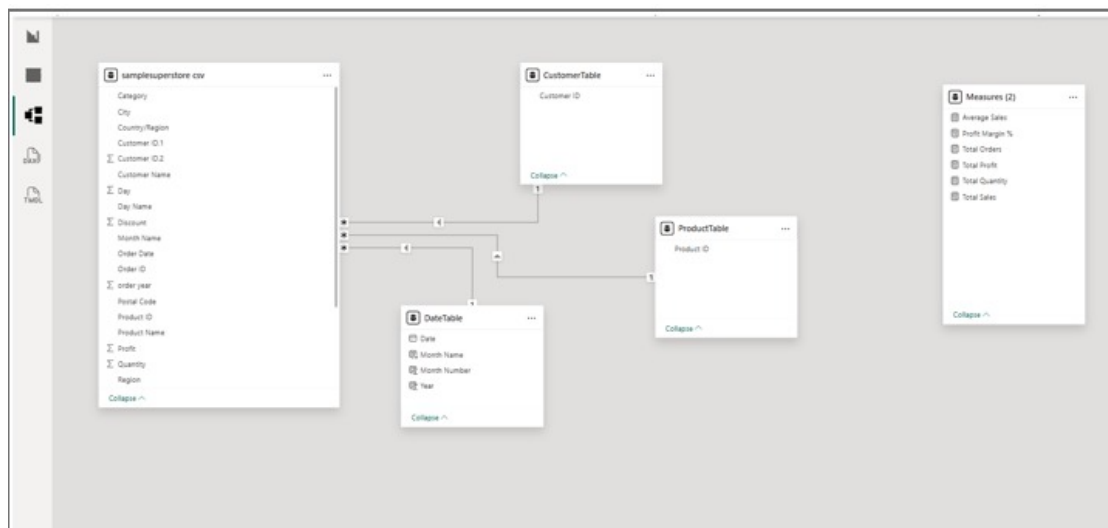
The dataset contains information about orders, customers, products, sales, quantity, profit, category, region, and segment.

Important columns: Order Date, Customer ID, Product

Name, Category, Sales, Quantity, Profit, Region, and Segment.

4. Data Preparation

Power Query was used to check data types, organize columns, and prepare the dataset. Tables and relationships were created for proper analysis.



5. Date Table Creation

A Date Table was created with:

Date

Year

Month Name

Month Number

The screenshot shows a Power BI interface. On the left, a DAX table named 'DateTable' is defined with the following code:

```

1 DateTable =
2 CALENDAR(
3 MIN('samplesuperstore csv'[Order Date]),
4 MAX('samplesuperstore csv'[Order Date]))
5

```

The table 'DateTable' has four columns: 'Date', 'Year', 'Month Name', and 'Month Number'. The data shows dates from 03-01-2023 to 26-01-2023, all in the year 2023 and month of January.

On the right, the 'Data' pane is visible, showing a search bar and a list of data sources: Measures (2), CustomerTable, DateTable, Date, Month Name, Month Number, Year, ProductTable, and samplesuperstore csv.

Date	Year	Month Name	Month Number
03-01-2023 00:00:00	2023	January	1
04-01-2023 00:00:00	2023	January	1
05-01-2023 00:00:00	2023	January	1
06-01-2023 00:00:00	2023	January	1
07-01-2023 00:00:00	2023	January	1
08-01-2023 00:00:00	2023	January	1
09-01-2023 00:00:00	2023	January	1
10-01-2023 00:00:00	2023	January	1
11-01-2023 00:00:00	2023	January	1
12-01-2023 00:00:00	2023	January	1
13-01-2023 00:00:00	2023	January	1
14-01-2023 00:00:00	2023	January	1
15-01-2023 00:00:00	2023	January	1
16-01-2023 00:00:00	2023	January	1
17-01-2023 00:00:00	2023	January	1
18-01-2023 00:00:00	2023	January	1
19-01-2023 00:00:00	2023	January	1
20-01-2023 00:00:00	2023	January	1
21-01-2023 00:00:00	2023	January	1
22-01-2023 00:00:00	2023	January	1
23-01-2023 00:00:00	2023	January	1
24-01-2023 00:00:00	2023	January	1
25-01-2023 00:00:00	2023	January	1
26-01-2023 00:00:00	2023	January	1

It was connected with the Order Date field for time-based analysis.

6. DAX Measures

The following measures were created:

Total Sales

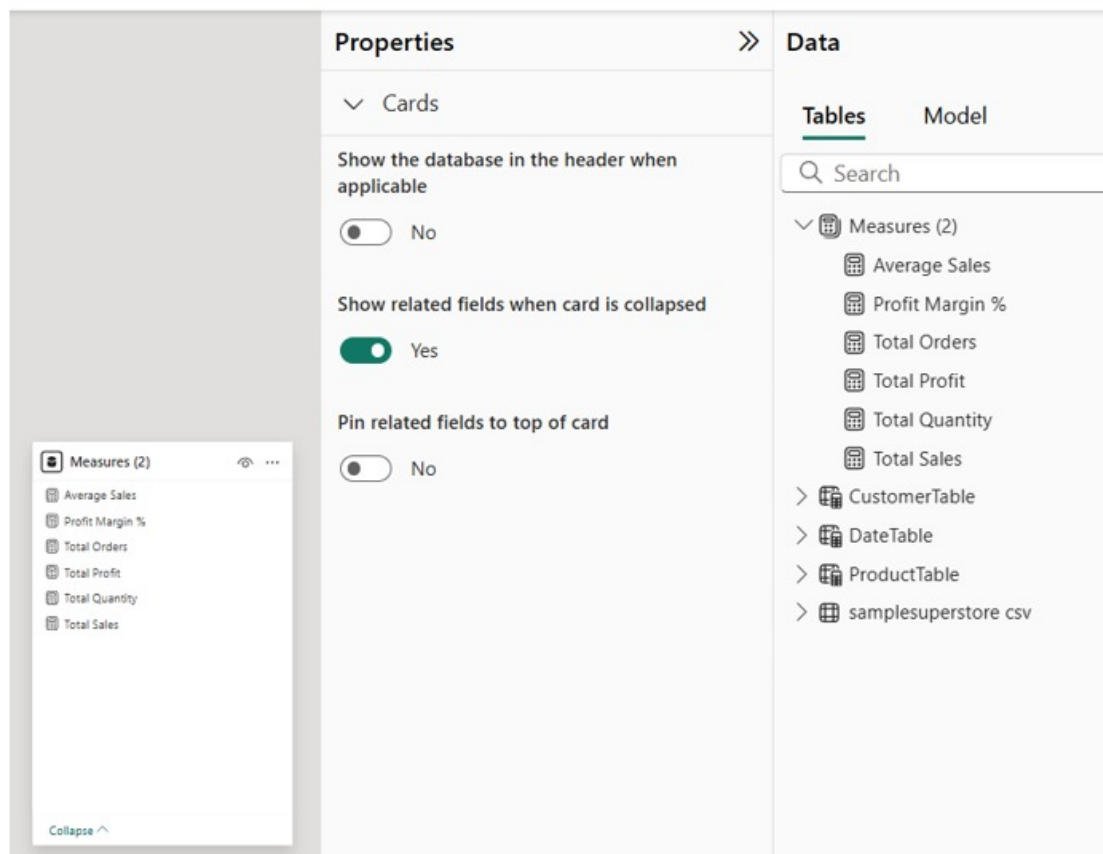
Total Profit

Total Orders

Total Quantity

Average Sales

Profit Margin %



7. Dashboard Components

The dashboard contains:

KPI Cards

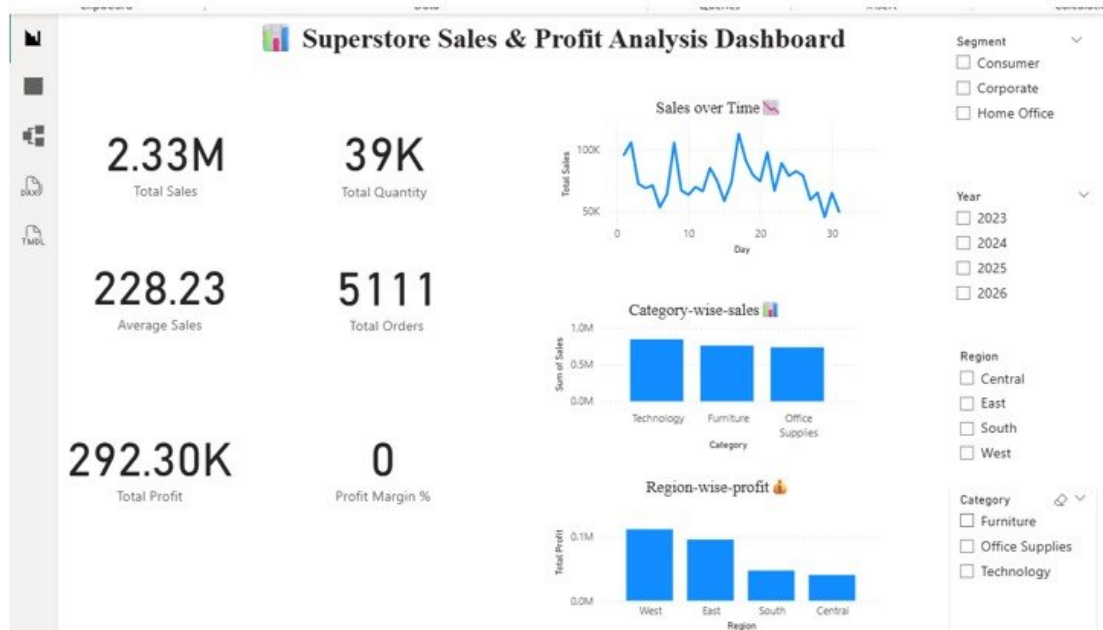
Category-wise Sales Chart

Region-wise Profit Chart

Sales Trend Line Chart

Slicers

Decomposition Tree



8. Key Insights

The dashboard helps analyze overall sales, profitability, product categories, regional performance, customer segments, and sales trends over time.

9. Advantages

Interactive analysis

Easy data visualization

Quick comparison of sales and profit

Flexible filtering

Time-based analysis

10. Conclusion

The project demonstrates how Power BI can be used for data preparation, modeling, DAX calculations, visualization, and interactive business analysis using the Superstore Sales Dataset.

11. Tools Used

Microsoft Power BI, Power Query, DAX, KPI Cards, Charts, Slicers, and Decomposition Tree.